Relational Algebra

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Overview

Content

- Selection
- Projection
- Product
- Join
- Union, Intersection, Difference
- Rename

Learning Outcomes

- Understand the meaning of each operator.
- Write relational algebra to query data in given relations.
- Calculate the data specified by a particular relational algebra.

Relational Algebra

What is the Relational Algebra?

Data Manipulation:

English $\Leftrightarrow$ Relational Algebra $\Leftrightarrow$ SQL queries

University

uID	uName	Country	Enrollment
...	...	...	...

- **English:** "Find all universities with more than 20000 students."
- **Relational Algebra:** $\pi_{uName}(\sigma_{Enrollment} > 20000(University))$
- **SQL:** SELECT uName FROM University WHERE Enrollment $>$ 20000

Relational Model and Relational Algebra

- **Relational Model:** Data $\rightarrow$ Relations
- Data Manipulation $\rightarrow$ operations on relations
- **Relational Algebra:**
  - A theoretical language with operations that work on relations.
  - Takes relations as input and produce new relations.
  - Operations won't affect the original relations!
  - Theoretical foundation for SQL.
- **Operators:** $+$,-,$\times$,$\div$ for numbers, &, |, $\neg$ for boolean
  - Common to Set-theoretic one
  - Specific to relations

Unary Operations

Example: University Applications

- What are the primary keys?

University		
uName	County	Enrollment
UON	Nott/shire	18000
CAM	Cam/shire	22000
UCL	Great/Lon	20000

Student			
SID	sName	GPA	HS
0135	John	18.5	100
0025	Mary	19.3	1000
0423	Mary	17.5	300

Apply			
SID	uName	Subj	Dec
0135	CAM	CS	'A'
0135	UON	CS	'A'
0423	UON	ENG	'R'

Example: University Applications

- Primary keys are:
  - **uName** for University.
  - **SID** for Student.
  - **(SID, uName, Subj)** for Apply.
- I want to know the informtion of the students who has a GPA less then 19.
  - What operator should I use?
  - How to write a query?

University		
uName	County	Enrollment
UON	Nott/shire	18000
CAM	Cam/shire	22000
UCL	Great/Lon	20000

Student			
SID	sName	GPA	HS
0135	John	18.5	100
0025	Mary	19.3	1000
0423	Mary	17.5	300

Apply			
SID	uName	Subj	Dec
0135	CAM	CS	'A'
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0423	UON	ENG	'R'

Relations: Mathematical definition

- A relation $\mathcal{R}$ of degree n, where the domains for each attributes are $D_1, \ldots, D_n$, is a subset of the Cartesian product of the domains:

$$\mathcal{R} \subseteq D_1 \times \cdots \times D_n$$

- Cartesian product:

$$D_1 \times \cdots \times D_n = \{(v_1, \ldots, v_n) | v_i \in D_i\}$$

- Example:
  - if $A_1 = \{1, 2\}$ and $A_2 = \{3, 4\}$, then
  - $A_1 \times A_2 = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$

Selection

- Selection works as filters.
- Let $\mathcal{R}$ be a relation with n columns and α is some properties of tuples.
- Selection from $\mathcal{R}$ subject to condition α is defined as:

$$\sigma_\alpha(\mathcal{R}) = \{(a_1, \ldots, a_n) | (a_1, \ldots, a_n) \in \mathcal{R}, \alpha(a_1, \ldots, a_n)\}$$

- σ: the selection operator.
- $(a_1, \ldots, a_n) \in \mathcal{R}$ means $(a_1, \ldots, a_n)$ is a tuple in relation $\mathcal{R}$.

What are properties?

- **Properties** are **expressions** connected with logical symbols, i.e., and, or, not.
- Each **expression** is either:
  - Attributes comparisons $(=, \neq, >, <, \geq, \leq)$
    - e.g., Enrollment $>$ HS
  - Or comparision between an attribute and a value
    - e.g., GPA > 15
- $\sigma_{GPA<19}(Student)$

Student			
SID	sName	GPA	HS
0135	John	18.5	100
0025	Mary	19.3	1000
0423	Mary	17.5	300

Excercise: Selection

- Find out all students with **GPA** more than 19.
- Find out all students with **GPA** more than 18 and high school size less than 1000.
- Find out all applications to **University of Nottingham (UoN)** with subject **CS**.

University		
uName	County	Enrollment
UON	Nott/shire	18000
CAM	Cam/shire	22000
UCL	Great/Lon	20000

Student			
SID	sName	GPA	HS
0135	John	18.5	100
0025	Mary	19.3	1000
0423	Mary	17.5	300

Apply			
SID	uName	Subj	Dec
0135	CAM	CS	'A'
0135	UON	CS	'A'
0423	UON	ENG	'R'

Excercise: Selection

- Find out all students with **GPA** more than 19.

$$\sigma_{GPA>19}(Student)$$

- Find out all students with **GPA** more than 18 and high school size less than 1000.

$$\sigma_{GPA>18\ and\ HS<1000}(Student)$$

- Find out all applications to **University of Nottingham (UoN)** with subject **CS**.

$$\sigma_{uName='UON'\ and\ Subj='CS'}(Apply)$$

$$\sigma_{uName='UON'}(\sigma_{Subj='CS'}(Apply))$$

Projection

- Projection works as slicing.
- Let $\mathcal{R}$ be a relation with n columns and $\mathcal{X}$ is a set of attributes.
- Projection of $\mathcal{R}$ on $\mathcal{X}$ is defined as:

$$\pi_{\mathcal{X}}(R)$$

- π: the projection operator.
- $\pi_{\mathcal{X}}(R)$ generate a new relation which only contains attributes from X.

Student			
SID	sName	GPA	HS
0135	John	18.5	100
0025	Mary	19.3	1000
0423	Mary	17.5	300

Excercise: Projection

- Get IDs and decisions from all applications.
- Get IDs and names of students with GPA greater than 19.

University		
uName	County	Enrollment
UON	Nott/shire	18000
CAM	Cam/shire	22000
UCL	Great/Lon	20000

Student			
SID	sName	GPA	HS
0135	John	18.5	100
0025	Mary	19.3	1000
0423	Mary	17.5	300

Apply			
SID	uName	Subj	Dec
0135	CAM	CS	'A'
0135	UON	CS	'A'
0423	UON	ENG	'R'

Excercise: Projection

- Get IDs and decisions from all applications.

$$\pi_{SID,Dec}(Apply)$$

- Get IDs and names of students with GPA greater than 19.

$$\pi_{SID,sName}(\sigma_{GPA>19}(Student))$$

- Can we change the order?

$$\sigma_{GPA>19}(\pi_{SID,sName}(Student))$$

Set Operations

Union

- The definition from standard set-theory:

$$A \cup B = \{x | x \in A \, or \, x \in B\}$$

- E.g., $\{a, b, c\} \cup \{a, d, e\} = \{a, b, c, d, e\}$

ID	Name
M139	John Smith
A368	Jane Brown
A367	David Jones

ID	Name
M140	Mary Jones
A222	Mark Brown
A367	David Jones

ID	Name
M139	John Smith
M140	Mary Jones
A368	Jane Brown
A222	Mark Brown
A367	David Jones

Union-compatible

- Two relations $\mathcal{R}_1$ and $\mathcal{R}_2$ are union-compatible if and only if they have the same number of attributes, and corresponding attributes have the same domain.

ID	Name
M139	John Smith
A368	Jane Brown
A367	David Jones

ID	Age
M140	23
A222	31
A367	28

Set Difference

- The definition from standard set-theory:

$$A - B = \{x | x \in A \, and \, x \notin B\}$$

- E.g., $\{a, b, c\} - \{a, d, e\} = \{b, c\}$
- Require union-compatible.

ID	Name
M139	John Smith
A368	Jane Brown
A367	David Jones

ID	Name
M140	Mary Jones
A222	Mark Brown
A367	David Jones

ID	Name
M139	John Smith
A368	Jane Brown

Intersection

- The definition from standard set-theory:

$$A \cap B = \{x | x \in A \, and \, x \in B\}$$

- E.g., $\{a, b, c\} \cap \{a, d, e\} = \{a\}$
- Require union-compatible.

ID	Name
M139	John Smith
A368	Jane Brown
A367	David Jones

ID	Name
M140	Mary Jones
A222	Mark Brown
A367	David Jones

ID	Name
A367	David Jones

Cartesian Product

- The definition from standard set-theory:

$$A \times B = \{(x, y) | x \in A, x \in B\}$$

- E.g., $\{a, b\} \times \{d, e\} = \{(a, d), (a, e), (b, d), (b, e)\}$
- Does not require union-compatible.
- Extended Cartesian product:

$$A \times B = \{(c_1, \ldots, c_n, d_1, \ldots, d_m) | (c_1, \ldots, c_n) \in A, (d_1, \ldots, d_m) \in B\}$$

Student			
SID	sName	GPA	HS
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0423	Mary	17.5	300

Apply			
SID	uName	Subj	Dec
0135	CAM	CS	'A'
0135	UON	CS	'A'
0423	UON	ENG	'R'

Example: Cartesian Product

Student x Apply							
S.SID	sName	GPA	HS	A.SID	uName	Subj	Dec
0135	John	18.5	100	0135	CAM	CS	'A'
0135	John	18.5	100	0135	UON	CS	'A'
0135	John	18.5	100	0423	UON	ENG	'R'
0025	Mary	19.3	1000	0135	CAM	CS	'A'
0025	Mary	19.3	1000	0135	UON	CS	'A'
0025	Mary	19.3	1000	0423	UON	ENG	'R'
0423	Mary	17.5	300	0135	CAM	CS	'A'
0423	Mary	17.5	300	0135	UON	CS	'A'
0423	Mary	17.5	300	0423	UON	ENG	'R'

- What do the tuples in red mean?
- How to solve the problem?

Join Operators

Natural Join Operator

- Student $\bowtie$ Apply (bowtie)
  - Similar to Cartesian Product but enforce equality on all attributes with the same name (SID in the previous case).
  - Automatically sets values equal when attribute names are the same.
  - Get rid of multiple copies of the attributes with the same name.

Student x Apply						
SID	sName	GPA	HS	uName	Subj	Dec
0135	John	18.5	100	CAM	CS	'A'
0135	John	18.5	100	UON	CS	'A'
0423	Mary	17.5	300	UON	ENG	'R'

Excercise: Natural Join

- Get names and GPAs of the students with high school size greater than 1000 who applied to CS and were rejected.
- Get names and GPAs of the students with high school size greater than 1000 who applied to CS at Universities with enrollment greater than 20000 and were rejected.

University		
uName	County	Enrollment
UON	Nott/shire	18000
CAM	Cam/shire	22000
UCL	Great/Lon	20000

Student			
SID	sName	GPA	HS
0135	John	18.5	100
0025	Mary	19.3	1000
0423	Mary	17.5	300

Apply			
SID	uName	Subj	Dec
0135	CAM	CS	'A'
0135	UON	CS	'A'
0423	UON	ENG	'R'

Excercise: Natural Join

- Get names and GPAs of the students with high school size greater than 1000 who applied to CS and were rejected.

$$\pi_{GPA,sName}(\sigma_{HS>1000\ and\ Subj='CS'\ and\ Dec='R'}(Student \bowtie Apply))$$

- Get names and IDs of the students who applied to CS at Universities with enrollment greater than 20000.

$$\pi_{sID,sName}(\sigma_{Subj='CS'\ and\ Enrollment>20000}(Student \bowtie Apply \bowtie University))$$

Theta Join Operator

- Cartesian Product satisfying certain properties.
- Can be implemented via Cartesian Product and Select operator.
- If $\mathcal{R}_1$ and $\mathcal{R}_2$ are two relations, θ is a property or properties, then the Theta Join Operator is defined as:

$$\mathcal{R}_1 \bowtie_\theta \mathcal{R}_2 = \sigma_\theta(\mathcal{R}_1 \times \mathcal{R}_2)$$

- The results consists of all combinations of tuples in $\mathcal{R}_1$ and $\mathcal{R}_2$ that satisfy properties θ.
- Can we define Natural Join based on Theta Join?

Rename Operator

- The rename operator has 3 forms. Suppose E is a relational algebra that generates a new relation, $S(T_1, \ldots, T_n)$.
  - $\rho_{R(A_1,\ldots,A_n)}(E)$: return a relation $R(A_1, \ldots, A_n)$
  - $\rho_R(E)$: returns a relation $R(T_1, \ldots, T_n)$
  - $\rho_{(A_1,\ldots,A_n)}(E)$: returns a relation $S(A_1, \ldots, A_n)$
- R is the new name of the relation generated by E
- $A_1, \ldots, A_n$ are the new names for the attributes in the new relation.

Rename Operator

Why we want a rename operator?

- Unifies schemas for Union, Difference and Intersection
  - List all Student and University names

$$\rho_{(name)}(\pi_{sName}(Student)) \cup \rho_{(name)}(\pi_{uName}(University))$$

- Help to disambiguation in self joins
  - Pairs of Universities in the same County.
  - We want to use rename and natural join

$$\sigma_{n1>n2}(\rho_{U_1(n_1,c,e_1)}(University) \bowtie \rho_{U_2(n_2,c,e_2)}(University))$$